

Diriyah Gate Company
Limited
a Single Shareholder
Limited Liability Company
CR: 1010648801



شركة بوابة الدرعية المحدودة
شركة شخص واحد
شركة ذات مسؤولية محدودة
السجل التجاري: ١٠١٠٦٤٨٨٠١

Schedule of Project Requirements

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Diriyah Gate Company Limited (DGCL)
DD-2023-118

DG II – Infrastructure Package 1
June 2, 2023





Schedule of Project Requirements

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Acronyms and Abbreviations

AASHTO	American Association for State Highway and Transportation
ASTM	American Society for Testing and Materials
ANSI	American National Standards Institute
AWWA	American Water Works
BS EN	British Standard European Norm
CIRIA	Construction Industry Research and Information Association
CITC	Communication and Information Technology Commission, KSA
DGCL	Diriyah Gate Company Limited
DIN	Deutsches Institut für Normung e.V.
DIN	EN Deutsches Institut für Normung e.V. European Norm
HDM	Highway Design Manual, KSA
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IES	Integrated Energy Solutions
IFC	International Fire Code
HDPE	High Density Poly Ethylene
ISO	International Standards Organization
MEWA	Ministry of Environment, Water and Agriculture
MOWE	Ministry of Water and Electricity
MOC	Ministry of Communications, KSA
MOH	Ministry of Housing
MOMRA	Ministry of Municipality and Rural Affairs
MOLSD	Ministry of Labour and Social Development
MOT	Ministry of Transport, KSA
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
NWC	National Water Company
RCRC	Royal Commission for Riyadh City
SASO	Saudi Standards, Metrology and Quality Organization
SIO	Saudi Irrigation Organization
SBC	Saudi Building Code
SEC	Saudi Electricity Company
LTI	Lost Time Injury

1. GENERAL

The tender document including the tender drawings and technical specifications is produced directly from Preliminary Design stage, prior to the development of detailed design deliverables. This has been done in order to expedite the tender process in line with the wishes, objectives and direction of the Employer, Diriyah Gate Company Limited (DGCL). As a result, this Tender may have some gaps in details of certain roads and infrastructure works and their associated drawings and technical specifications, which will be filled-in with the issuance of tender addenda during tender floatation period based on the outcome of the detailed design deliverables. Therefore, the bidders are expected to fully understand the works comprehensively going through the conditions of contract, scope of works, tender drawings, technical specifications, particular specifications, BOQ, etc.

2. INTRODUCTION

The Diriyah Gate Phase II (DGII) project site is located to the west of the city of Riyadh, Kingdom of Saudi Arabia. It is about 25km from King Khalid International Airport. The DG II site covers a land area of about 630 Ha, surrounded by two major corridors. These are the King Khalid Road (KKR), which forms a physical divider for DGII Master Plan in two sections (RHS and LHS) and the Western Ring Road (WRR), creates a physical division between DGI and DGII.

The site has been divided into three packages, which have generally been used as the basis for design and construction packages with typical designation 'Package 1', 'Package 2' and 'Package 3' dependent on the package of the development.

This section provides a brief statement of the works to be carried out under Package 1 and provides a Schedule of Project Requirements. Where requirements within this section contradict those contained within other sections of the tender documentation, the requirements contained herein shall take precedence. The hierarchy structure of sections contained within this tender document is provided below.

- A. Schedule of Project Requirements:
 - a. This document, which also includes scope of works
- B. General Specification:
 - a. The latest versions of general specifications of local authorities and international standards to be followed as presented in Table 2 of this document
- C. Particular Specifications:
 - a. The following works have a separate and standalone document specifications:
 - i. Pumping stations
 - ii. MV culverts

3. CONSTRAINTS

The following Table 1 contains a high-level summary of constraints and issues on the site that the Contractor should account for within his bidding documentation. The Contractor will be required to develop and execute appropriate plans of action to manage and mitigate these issues and others that impinge on delivery of the scope. This is a non-exhaustive list. It is required that the Contractor, during the bidding phase, familiarize themselves with the site and address their role as delivery agent of the scope of works in a safe, efficient, timely, cost-effective manner that achieves the necessary standards of quality and care.

Table 3-1: Project Constraints Summary

Item	Title	Description
1.	VVIP occupants of DGII	There are a number of high profile VVIP residents within DGII. The Contractor will be required to liaise and coordinate with these VVIP's to minimize disruption. The liaison with VVIP's will mainly be through DGCL.
2.	Liaison with residents and government authorities	Generally, the liaison with VVIP's as well as other residents will be through DGCL.
3.	Liaison with government authorities and service providers.	Generally, the Contractor shall liaise with, in coordination with the Employer, Government Authorities and Service Providers for the execution of the Works.
4.	Maintenance of services to existing residents	All existing residents will require their services to be maintained. This includes potable, irrigation, power, telecoms, etc. Appropriate temporary works and diversion plans will need to be prepared well in advance for agreement with DGCL and the VVIP's.
5.	Maintenance of access	Access to existing residents as well as other contractors and users of DGII will need to be maintained. Appropriate temporary works and diversion plans will need to be prepared well in advance for agreement with DGCL and the VVIP's as well as other residents.
6.	Other contractors on site	This contract is one of several that DGCL are procuring. The Contractor will be required to ensure full cooperation and collaboration to the works so that safety is maintained as well as free flow of men and materials throughout the site. Refer to Section 6.2 of this document for further details.
7.	Authority approvals	The Contractor shall be responsible for submitting and obtaining the necessary approvals from the authorities in coordination with DGCL. The Contractor will be required to seek the necessary NOC's from authorities (e.g. Civil Defense, SEC, CITC, NWC, SOI, etc.).
8.	Geotechnical conditions and ground water	Reference is made to the geotechnical survey completed by DGCL. The geotechnical conditions vary across the site and consists of soils that are prone to weathering. DGCL require minimizing damage to escarpments – which are poorly compacted/weathered soil and sandstone.

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Item	Title	Description
9.	Work Permits	Contractor shall be responsible for, in coordination with the Employer, submitting and obtaining all required Work Permits prior to commencement of the works.
10.	Escarpment Protection and other areas outside the limits of the Works.	The Contractor shall be responsible for any damage caused by his operations in the delivery of the Works at his own cost.

4. LOCATION

The DGII project site is located to the west of the city of Riyadh, Kingdom of Saudi Arabia. The DG II site covers a land area of about 630 Ha, surrounded by two major corridors i.e. KKR and WRR. The construction infrastructure and utility works of the project is divided into three packages as shown below. This tender is for Infrastructure Package 1 only.

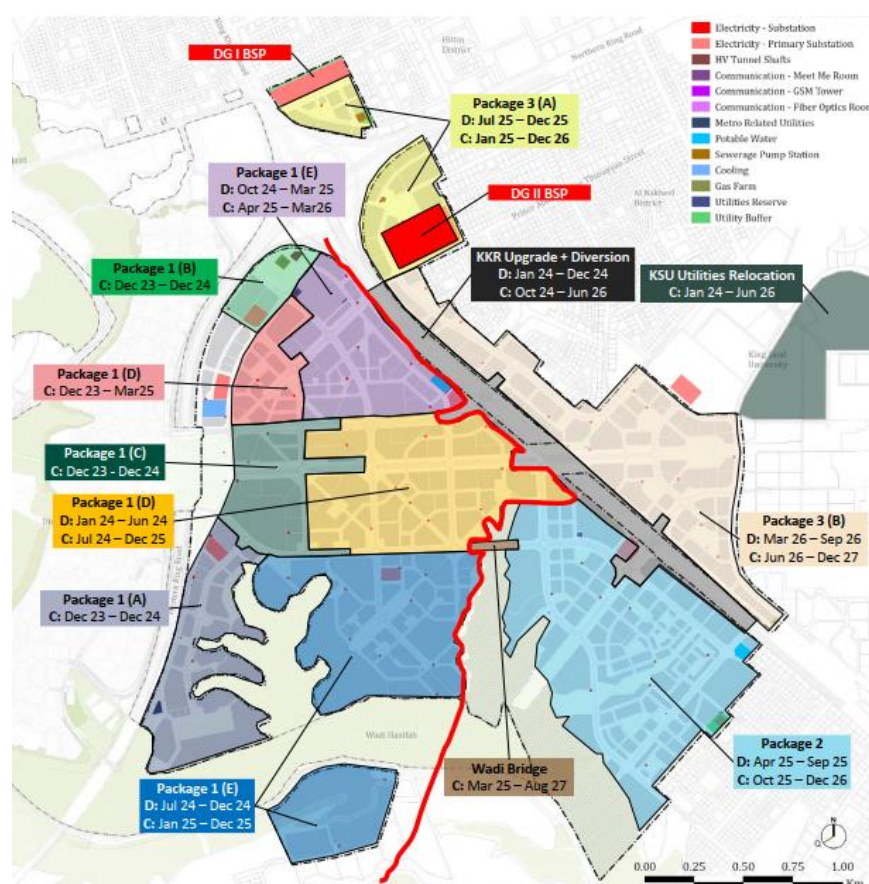


Figure 4-1: Location of Packages within DGII

A. Package 1: Central Site (This Tender)

Covers a land area of 322 hectares and is located in the center of the development, bounded by the WRR and KKR but excluding the area to the east of East Wadi.

B. Package 2: South-East Area

Covers an additional land area of 167 hectares and is located to the east of Package 1 and in-between the East Wadi and KKR.

C. Package 3: Eastern Area

Covers an additional land area of 123 hectares and is located to the east of KKR.

5. GROUND CONDITIONS AND CONDITION SURVEYS

The following ground investigation data (site-wide) will be provided by the Employer.

- A. ACTS, Geotechnical Interpretative Report MISK City Development (Area 1), Riyadh, KSA, ACTS-Q-8097/Rev 01, May 2019.
- B. ACTS, Geotechnical Interpretative Report MISK City Development (Area 2), Riyadh, KSA, ACTS-Q-8097/2/Rev 00, May 2019.
- C. Soil and Foundation Co. Ltd (SAFCO), A Geotechnical Investigation Report for Diriyah Gate Development Project Al-Darriyah, Riyadh, KSA, RSM 15492-Rev 0, June 2019
- D. ACES, Diriyah Gate II Detailed Master Plan Design, Geotechnical Factual Report, RPT. No. S22000020-Rev 2, November 2022.

The Contractor shall make his own judgement and assessment on the ground conditions of the site checking the geotechnical investigation documents and undertaking site visits.

The Contractor shall be responsible for assessing the adequacy, correctness and comprehensiveness of the information and undertaking any additional survey which he deems to be necessary.

The Contractor shall be responsible for seeking any additional data, which may be available, and for undertaking any additional geotechnical or other investigations and/or surveys that may be necessary for the Works.

The Contractor shall carry out a visual and photographic condition survey of all elements of the existing services, facilities and adjacent areas not being demolished, or as otherwise required by the Engineer, within Fourteen days of the Commencement Date and prior to the commencement of any demolition or construction work. The Contractor will also establish the levels of these areas.

The Contractor shall undertake all the necessary requirements to identify any existing services that are/potential to be impacted by the associated works in the Contract.

Where works are to be undertaken in areas containing existing services, the Contractor shall establish their precise location by hand-digging trenches or by other suitable means such as electronic scanning before laying any adjacent new service. The Contractor shall provide such information regarding existing services to the Engineer and shall include such details on his record and as-built drawings.

The Contractor shall be responsible for the financial cost for the physical repair, reinstatement, or replacement of any of the existing facilities, services, drainage, and adjacent areas damaged as a result of his activities. Any such repair, reinstatement or replacement shall be undertaken by the Contractor to the satisfaction of the Engineer and relevant third parties.

6. SCOPE OF WORKS

This tender package generally includes the primary infrastructure within Package 1 including some works within and surrounding this area that tie into the development.

The scope of works comprises the following and shall be undertaken in accordance with the issued tender drawings and any additional instructions/direction from the Engineer.

Particular specifications (where provided) shall take precedence over general specifications unless otherwise instructed.

6.1 General

Package 1 includes the construction, testing and commissioning, and handing over of all infrastructure, earthworks grading, roads, utilities, utility facilities, and any supporting temporary works.

General requirements include:

- A. Maintenance of site office equipment and facilities as stated in Section 7 of this document and the contract documents for the duration of contract.
- B. Obtaining all required permits and approvals from the Employer and the authorities having jurisdiction.
- C. All temporary works required for completion of works including but not limited to dewatering, service diversions, and maintenance of temporary access to residents, contractors, and service providers and others as required by the Engineer and Employer.
- D. The Contractor will prepare and submit detailed, well-coordinated shop drawings, combined and for each separate utility, for the approval of the Engineer and the relevant Departments and Authorities.
- E. It shall be a requirement of the contract that each existing utility shall remain fully operational until such time as the proposed replacement or diversion is in place, commissioned and ready for jointing at connection points, in coordination and in accordance with the requirements of the Owner of the Utility.
- F. Preparation of as-built drawings in accordance with the Employer requirements and as well as authority and service provider standards (where applicable).
- G. The Contractor and upon completing the work shall provide the Employer with three original sets of As-Built drawings accurately reflecting the executed works and installation details.
- H. General requirements shall conform to the approving authority (Employer and pertinent Authorities having jurisdiction) General Specifications where not otherwise stated or where conflicting requirements are stated amongst other authority specifications. Where there is contradiction with the General and Particular Specifications, the Particular Specifications shall take precedence.
- I. Any necessary works external to Package 1 limits of works to connect new infrastructure (roads and services) to the existing infrastructure. Including any necessary demolition of existing roads, utilities and

services to enable tie-in. This shall include coordination with and approval from DGCL, residents/ users, authorities and service providers to pre-prepare and carry out any required activities to tie in.

- J. Any necessary works external to Package 1 limit of works to connect new infrastructure (roads and services) to the infrastructure of Package 2 and Package 3 at and/or local to the boundary between the packages. This shall involve coordination with the Contractors for Package 2 and Package 3 and all other Contractors as listed in Section 6.2 of this document and could involve work outside of the limits of Package 1 such as coordination with residents and plot owners within the overall site.
- K. The Contractors attention is drawn to the fact that he will not have exclusive access to the whole of the limit of Package 1 and that therefore shall liaise with the other contractors working within the limit of works to maintain any required construction access and interfacing during the contract period. There may be other contractors working within the limit of works as listed in Section 6.2 of this document.
- L. The Contractor shall upon completing the works provide the Employer with detailed Operation and Maintenance Manuals (With applicable Arabic translations) in three originals. The manuals shall include but not limited to:
 - a. Complete and marked literature of all products used on the project and its reordering part numbers.
 - b. List of the recommended spare parts for a period of 5 years
 - c. Original print of the As Built drawings in addition to all relevant editable copies in CAD, PDF, GIS and BIM formats
- M. The Contractor shall make himself fully informed of the Environmental authorities' requirements, which need to be met to complete the construction work.
- N. The Contractor shall comply with findings of Diriyah Gate II Environmental Impact Assessment (EIA), and other environmental & sustainability requirements as listed in Section 6.16 of this document.
- O. The Contractor shall note that the project is located within near proximity to number of existing facilities and properties as well as other construction projects. Tender submissions and proposed rates shall reflect all necessary coordination, cooperation, permitting and any requirements and constraints to the Works as may apply.
- P. The Contractor shall be responsible for developing a comprehensive construction phasing and temporary traffic management plan to meet the requirements included in the tender documents and shall make full allowance for all temporary traffic management required to satisfy these requirements in their tender price and programme.
- Q. The Contractor shall be aware of the requirements to maintain all public roads open to traffic at all times, unless otherwise agreed with the Engineer, with the minimum number of lanes maintained in each direction agreed with the Engineer. These requirements may place constraints on the phasing of construction and must be reflected in the construction program.
- R. The Contractor shall, in addition, note that, specified materials that do not meet the stated specifications will not be permitted under any circumstances.

- S. The Contractor shall be aware of the strict programme durations and shall account for the provision of sufficient resources to ensure delivery.
- T. The Contractor shall make himself fully informed and familiar with the surroundings by way of site visits, site inspections, collecting information or any other means deemed necessary and take into consideration the local situation with regard to availability of construction materials/suppliers, volatility and escalation of prices, difficulties associated with bringing and accommodating staff/labourers and obtaining lands for Site Establishment.
- U. Due to the extent of new, diverted, replaced and protected underground services, the Contractor shall note the requirement to carry out an adequate number of utility trial trenches / pits to identify the exact location and level of all existing underground services and ascertain in advance of construction, the route, level, and appropriate execution sequence of all proposed underground networks along their entire lengths, resolving all clashes at site and taking the traffic diversions into consideration.

The DG II Package 1 infrastructure includes single and dual carriageway roads, earthworks & grading, wet and dry utility networks and supporting infrastructure. The infrastructure includes but is not limited to the following:

Table 6-1: Scope of Works (General)

Scope of Works (General)	
Storm water networks	Roads
Potable water networks	Pedestrian underpasses / Box Structures
Pumping stations	Street lighting networks
Irrigation networks	Traffic signals
Wastewater networks	CCTV
MV & LV electrical systems	GSM provision
Statutory signage	Security
ICT systems	Fire & Life Safety systems
Telecommunications networks	Firefighting including fire detection and suppression systems
Gas networks	

The Contractor shall be responsible for undertaking all construction related activities, from site set up, through the construction phase, testing & commissioning, handover and close out.

6.2 List of 3rd Party Works for Coordination

There will be several works undertaken by other parties outside of the scope of the Contract (within or adjacent to the project boundary). The Contractor shall be fully responsible for all; interfacing, coordination and accommodating for the following but is not limited to:

- A. DG I Infrastructure Contractors

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- B. DG II Infrastructure Contractors
- C. DG II Enabling Works Contractor
- D. DG II Public Realm Contractors
- E. DG II Building / Asset Contractors
- F. WRR Contractor
- G. KKR Contractor
- H. SEC Contractors
- I. Utility and Services Demolition/Diversion/Relocation Contractors

6.3 Standards & Specifications

The general requirement is that the works are carried out in accordance with the latest version of the associated Saudi Arabian specification.

Table 6-2: Reference Specifications for the Works

Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
General - Project laboratory requirement - Construction and maintenance of detours - Quality control of materials - Storage and handling of material	General Specifications for Road and Bridges Construction – Part-1 – General	Ministry of Transport (MOT)	
Earthwork and Site Clearance	General Specifications for Road and Bridges Construction – Part-2 – Earthwork	Ministry of Transport (MOT)	
Piling	General Specifications for Road and Bridges Construction – Part-5 – Concrete, Steel and Structure	Ministry of Transport (MOT)	- MOMRA - AASHTO LRFD Bridge Design Specifications - The ICE Specification for piling and embedded retaining walls - BS 8004:2015 Code of practice for foundations

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Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
Potable water & Firefighting Network	Standard Technical Specifications for Water Supply Networks	- National Water Company (NWC) for network - Civil Defence Department (for fire hydrants locations & spacing)	- MEWA - MOWE - AWWA - MOMRA - MOH - IFC - SASO - ASTM - ISO - BS EN - DIN - General Directorate of Civil Defence – Fire Fighting and Requirements for protection from fire - Saudi Building Code SBC 2018 - National Fire Protection Association (NFPA) 1, 22, 24, 25, 1142 - ANSI/NSF Standard 60 & 61 (materials shall comply)
Wastewater Network	Standard Technical Specifications for Water Supply Networks	- National Water Company (NWC) for network - Civil Defence Department (for fire hydrants locations & spacing)	- MEWA - MOWE - AWWA - MOMRA - MOH - IFC - SASO - ASTM - ISO - BS EN - DIN - General Directorate of Civil Defence – Fire Fighting and Requirements for protection from fire - Saudi Building Code SBC 2018

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Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
			- National Fire Protection Association (NFPA) 1, 22, 24, 25, 1142 - ANSI/NSF Standard 60 & 61 (materials shall comply)
Irrigation Network	- MOMRA- Specifications and design guidelines and the implementation of modern irrigation (a field of afforestation and public parks in the municipality). - Applicable specification in the kingdom of Saudi Arabia including RCRC and Riyadh Municipality /Amana governing specifications - Applicable specifications in the kingdom of Saudi Arabia including MEWA/NWC/SIO for main supply line from the source point	- Royal Commission for Riyadh City (RCRC) / Riyadh Municipality (Amana) for internal network. - MEWA/ NWC/ SIO (Saudi Irrigation Organization) for Main supply line from Source point up to DGCL scope area.	- MOMRA - SIO - MOT - MEWA/NWC - MOWE - MOH - SASO - BS EN - ISO - ASTM
Stormwater System	Applicable specification in the Kingdom of Saudi Arabia including RCRC, Riyadh Municipality /Amana	Royal Commission for Riyadh City (RCRC) / Riyadh Municipality (Amana)	- MOC/MOT - HDM - MOMRA - CIRIA - DMRB - SASO

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Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
	- and MOMRA governing specifications • MOC/MOT-General Specifications for Road and Bridges Construction • MOMRA, General specifications for building construction		• BS EN • AASHTO • ASTM • DIN EN • ISO • Jeddah Stormwater Design Manual by AECOM
Electrical Distribution	• DPS: Distribution Planning Standards • SDCS: SEC Distribution Construction Standards • SDMS: SEC Distribution Material Standards	• SEC	• IEC • BS EN • ASTM • DIN EN • NFPA • ANSI/IEEE • SASO • ISO • NEMA
Street Lighting	• General Specifications for Road and Bridges Construction – Part-1 – General • General Specifications for Road and Bridges Construction – Part-7 – Roadway Lighting and Traffic Signals	• Ministry of Transport (MOT)	• IES • NEC • IEC • NEMA • ANSI • BS EN • IEEE • SASO • MOMRA

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Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
Digital Infrastructure	- Local Telecom regulations; Telecom utility providers STC, Mobily and Zain - DGCL guidelines for security and communication	<u>CITC</u>	- TS 4111: HDPE main, mini & micro ducts - Saudi Building Code (201/302/303/304/306/401/501/601/801) - SASO Part 1,2 and 3
Roads	- General Specifications for Road and Bridge Construction 1998 - Highway Construction Manual - Highway Materials Manual - Addendum to the MOC November 1998 General Specification Parts 1,2,3 and 4 - Traffic Safety Manual 2010 - MOT Hot Mix Asphalt Design system manual - Special Specification for Road Marking 2016 - Road Sign Installation and Maintenance - KSA Work Zone Traffic Manual and Handbook	Ministry of Transport (MOT)	- ASTM - BS - BS EN
Bridges	General specifications for Road and Bridge Construction 1998	Ministry of Transport (MOT)	- MOMRA - AASHTO LRFD Bridge Design Specifications

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Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
Gas Network	• AMANA / municipality rules and regulations • Saudi Building Code (SBC) • Civil Defense Regulations • UAE ADNOC Gas Distribution Regulations and Guidelines • National Fire Protection Association (NFPA)	• AMANA • SBC • Civil Defense • ADNOC • NFPA	• Relevant International Standards such as: ○ ASME ○ ASTM ○ British Standards ○ EN standards ○ ISO Standards ○ Institute of Gas Engineers and Managers (IGEM).
Health, Safety & Environment	• Health and Safety Employer's Requirement – Volume 6, Schedule 9 • Royal Decree No. 51 of 2005 Labour Law; Part 8 Protection against Occupational hazards • Ministerial Decision No. 404 of 1974 First Aid Facilities at Work Sites • Ministerial Decree No. 2833 of 2006 Occupational Injury Reporting	• DGCL • MOLSD • MOH • MOT	

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Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
	- Ministerial Order No. 1399 of 2007 Worker Lodging - SAG Ministry of Interior Civil Defence construction regulations - KSA Work Zone Traffic Manual and Handbook		
Environment and Sustainability	- The General Authority of Meteorology and Environmental Protection (GAMEP), General Environmental Regulations and Rules for Implementation (GERRI), 2001 - International Finance Corporation (IFC) Performance Standards on Environmental and Social Sustainability - Requirements of LEED BD+C and Cities and	- DGCL - MEWA - NCEC - IFC, The World Bank Group - Leadership in Energy and Environmental Design, U.S. Green Building Council	Ministry of Environment, Water and Agriculture (MEWA), National Centre for Environmental Compliance (NCEC) standards and regulations applicable to infrastructure projects.

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Discipline	Specification Document Title	Specification Authoring Authority or Organization	Other Applicable Codes & Specifications
	Communities (as applicable)		

Note: The Contractor to confirm and adopt the latest version of all the Specifications and standards at the time of construction with prior approval from the Engineer.

6.4 Grading

- A. This section covers the performance of all the work and appurtenances required for the earthwork grading of road right of ways of the project as per the project drawings, and other contract documents.
- B. Removal, relocation and satisfactory disposal of all man-made features including but not limited to exiting road infrastructure, road signage, underground and over ground utilities boundary walls, fences, culverts, buildings, structures, abandoned services and pipelines, and any other obstructions which are not designated or permitted to remain in the road right of ways.
- C. The Contractor shall be responsible to collect the existing data of road infrastructure and other man-made features which are subjected to be demolished. The Contractor shall be responsible to submit the demolition plan and schedule to the Engineer and respective authorities having jurisdiction for approval before grading commencement.
- D. The Contractor shall perform the clearing and grubbing, of topsoil consisting mainly of loose soil, vegetation, and organic matters, drift sand, unsuitable soil and rubbish by scarifying the road right of ways area to a minimum depth of 150 mm from the natural ground level. All materials resulting from the above operations shall be removed from the site, loaded and transported and off-loaded, spread, and leveled to legal/ approved dump areas as directed by the Engineer and in compliance with the authority's rules and regulations.
- E. The excavation to formation of the road level contours, elevations, and dimensions are indicated on the road drawings. All suitable/ approved material removed by excavation shall be used as far as practicable in the formation of embankments or elsewhere when and as directed by the Engineer in compliance with the standards and specifications. It shall be the responsibility of the Contractor to handle excavation in any manner he sees fit, provided that suitable materials shall be available when required. No additional compensation shall be allowed for any special sequence of excavating or placing of such materials or any re-handling of materials.
- F. The material suitable for use as embankment shall be entirely imperishable and shall be determined to comply with the highway's materials manual of Ministry of Transport (MOT) as acceptable by the Engineer.
- G. Filling to formation level contours, elevations, and dimensions are indicated on the drawings and compaction of formation to the approved level of compaction in accordance with the MOT specifications.
- H. The compaction test of the embankment shall be performed as per the MOT standard and procedures of testing materials. The compaction trials shall comprise of field compaction and in-situ tests together with laboratory tests carried out on samples of the compacted fill. Samples taken for in-situ moisture content determination shall be taken without delay to a nominated laboratory for immediate testing.
- I. The Contractor shall obtain no-objection certificates (NOCs) and construction permits from the relevant authorities as required prior to commencing work.
- J. The Contractor shall follow the Ministry of Transport work zone manual for temporary traffic management to ensure the safe operations during the construction work.

- K. The Contractor shall ensure the site eco system and environmental characteristics are maintained during the construction in accordance with relevant regulatory authorities. In particular, the air quality of surrounding developments shall be monitored and maintained.

6.5 Roads

- A. The construction of roadworks includes two-way single carriageways and multiple dual carriageway roads.
- B. The scope of work includes the overall construction of road but excluding non-motorised user paths and landscape. Those will be constructed under a separate contract.
- C. The construction of road network includes all types of junctions, including traffic signalised junction and roundabouts, which fall within the project scope. Where it is necessary, the Contractor shall prepare the shop drawings to elaborate the overall design and specifically at junction and roundabouts subject to approval from the Engineer.
- D. The existing road pavement shall be demolished and reconstructed at tie in points as per standards and specifications in order deliver a synchronized seamless construction joint at interface with existing road network.
- E. The Contractor shall prepare the shop drawings at construction joints and Tie-in design with external networks at access points for authority approvals and other necessary construction permits.
- F. The Contractor shall provide the safety barriers, cross sectional elements and road side features shown in the project drawings.
- G. Prime and Tack coat to be provided at granular and bituminous base course layers respectively as per the project specifications.
- H. The Contractor shall furnish and install the road marking and signage on the road.
- I. Ministry of Transport (MOT) and Amanah Riyadh General Specifications shall be followed for the project execution. However, where there is a contradiction, MOT specifications shall take precedence.
- J. The surplus material and waste materials shall be disposed at locations as approved by the Engineer in compliance with Authority's Rules and Regulations.
- K. The Contractor shall follow the Ministry of Transport work zone manual for temporary traffic management to ensure the safe operations during the construction work.
- L. The Contractor shall ensure the site eco system and environmental characteristics are maintained during the construction in accordance with relevant regulatory authorities. In particular, the air quality of surrounding developments shall be monitored and maintained the project execution.

6.6 Electrical Distribution

- A. This section covers the performance of all the work and appurtenances required for electrical power distribution as per the project drawings and other contract documents.

- B. Final shop drawings.
- C. Preparation and obtaining approval of the shop drawings from the Engineer and Saudi Electricity Company (SEC)
- D. Approval for all the material from the Engineer and SEC.
- E. Excavation and bedding of trench on all type of soil, and backfilling with imported sand, and putting warning tags according to the SEC Specifications or as instructed by the Engineer.
- F. Medium Voltage road crossing duct bank, importing and installing plastic pipes (150mm) including concrete casting and burial according to SEC specification or as instructed by the Engineer.
- G. Importing and laying medium voltage cable including testing according to SEC specification or as instructed by the Engineer.
- H. Importing and laying low voltage cable including testing according to SEC specification and drawings or as instructed by the Engineer.
- I. Importing and installing medium voltage cable connectors, Aluminium box PVC/STA XLPE including testing according to SEC specification or as instructed by the Engineer
- J. Importing and installing low voltage cable connectors, including testing according to SEC specification.
- K. Importing and installing Medium Voltage cable endings, Copper box, XLPE PVC/STA, including testing according to SEC specification.
- L. Supply and install 50mm DIA HDPE duct in trenches for Fiber Optic Cable according to the contract drawings and documents and SEC specifications and drawings.
- M. Supply and install Fiber Optic cable pull pit as per SEC technical specification or as instructed by the Engineer
- N. Supply and install Fiber Optic Splicing Pit as per SEC technical specification or as instructed by the Engineer
- O. Supply, Install and test MV RMU for Normally-Open Points according to SEC specification or as instructed by the Engineer
- P. Supply and installation of all substation bases, (if required).
- Q. Connection to already constructed electrical conduit networks (if required).
- R. Disposal of surplus excavated material, waste materials, ducting and the like as directed by the Engineer in compliance with the Authority's Rules and Regulations.
- S. Testing and commissioning of all new electrical assets provided under this contract.
- T. Demolition works:

- a. Diversion, relocation, modification, or removal of already constructed networks where required as per the drawings.
- b. The Contractor shall ascertain:
 - i. The length of the MV cables
 - ii. The type of the MV cables
 - iii. The number of poles of the MV networks
 - iv. The number of the MV/LV substations
 - v. The length of the LV cables
 - vi. The several types of the LV cables
 - vii. The number of poles of the LV networks
 - viii. The number of the LV feeder pillars
- c. The Contractor shall be responsible for collecting information with regards to the existing power network from the relevant authorities.
- d. The Contractor shall be responsible to submit to the Engineer and respective authority the connection/disconnection plan and schedule for final approval before construction commencement.

6.7 Street Lighting

- A. This section covers the performance of all the work and appurtenances required for streetlight as per the project drawings and other contract documents.
- B. Final shop drawings and authority approvals for proposed street lighting.
- C. Submit material selection to the Employer for approval prior any construction commencement
- D. Supply and installation of street lighting substations as per MOT and SEC codes, specifications and drawings.
- E. Supply and installation of street lighting low voltage cables at any soil conditions as per specification and drawings
- F. Supply and install single core PVC insulated copper conductor cables for circuit protective conductors (CPC)/ECC
- G. Supply and installation of street lighting columns as per specifications and drawings and in accordance with authority standards and to the approval of the Employer.
- H. Supply and installation of street lighting luminaire as per specifications, drawings, and in accordance to authority standards and to the approval of the Employer.
- I. Supply and installation of columns foundation as per specifications and drawings and in accordance with authority standards

- J. Supply and installation of the earthing system as per specifications, drawings and in accordance to authority standards
- K. Excavation in all kind of soil and supply/installation of uPVC pipes for lighting cables as per specifications, drawings and in accordance with authority standards including backfilling and pull pipes
- L. Supply and installation of road crossing ducts as per specifications, drawings and in accordance with authority standards
- M. Supply and install of draw pits as per specifications, drawings and in accordance with authority standards
- N. Supply, installation and testing of street lighting LV feeder pillars as per specifications, drawings and in accordance with authority standards
- O. Termination of cables and energization and commissioning of lighting poles.
- P. Lead-in ducts are to be provided to each proposed street lighting pole location works where the poles are in ROW.
- Q. Lighting pole foundation to include 2 No ducts for power cables and 2 No ducts for other.
- R. Project particular specifications, (MOT and SEC) specifications shall be followed.
- S. Demolition works:
 - e. During production of this tender documents, there is an existing street lighting network from eastern entrance towards the middle of the site.
 - f. The Contractor shall ascertain:
 - i. The length of the LV street lighting cables
 - ii. The several types of the LV cables
 - iii. The number of poles of the street lighting networks
 - iv. The number of street lighting fixtures
 - v. The number of the street lighting LV feeder pillars
 - g. The Contractor shall be responsible for collecting the existing street lighting network from the relevant authorities.
 - h. The Contractor shall be responsible to submit to the Engineer and respective authority the connection/disconnection plan and schedule for approval before construction commencement.

6.8 Telecommunications

- A. This section covers the performance of all the work and appurtenances required for digital infrastructure (telecom) as per the project drawings and other contract documents.
- B. Final shop drawings and authority approvals for proposed digital infrastructure and GSM layouts.
- C. Collection of all materials provided from Authority/ service providers stores inclusive of delivery to site.

- D. Supply and installation of all ducting, including fittings and draw cords, as per the drawings (where not provided by Authorities/ service providers).
- E. Supply and installation of all chambers and FDT (where not provided by Authorities/ service providers).
- F. Connection to already constructed telecommunication containment networks.
- G. Connection to tie-in point of data center and plot connections. Infrastructure utility network shall be provided up to the plot boundaries.
- H. Provision of power, earthing protection system and telecom duct connection for GSM antenna plots and POPs.
- I. Diversion, relocation or removal of already constructed networks where required as per the drawings.
- J. Testing and commissioning of all the new telecommunication duct network including mandrel testing in accordance with Authority/ service providers requirements.
- K. Disposal of surplus excavated material, waste materials, ducting and the like as directed by the Engineer in compliance with Authority's Rules and Regulations.
- L. CITC and other General Specifications as mentioned in Section 3.1 shall be followed.
- M. The following works are NOT included in the works, however may require coordination:
 - a. Supply and installation of telecommunication cabling and equipment is not included in the scope of works.
 - b. Construction of Data center and associated MEP work is not included in scope of work for infrastructure Contractor.
 - c. Supply and install of GSM towers
- N. Demolition works:
 - a. The Contractor shall be responsible for collecting information with regards to the existing telecommunication network from the relevant authorities.
 - b. The Contractor shall be responsible to submit to the Engineer and respective authority the connection/disconnection plan and schedule for final approval before construction commencement.

6.9 Storm Water Drainage Systems

- A. This section covers the performance of all the works and appurtenances required for storm water drainage network and appurtenances as per the project drawings and other contract documents.

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- B. The works include construction of positive drainage network with pipe, manholes and catch basis, open channels, outfall structure, ponds, etc. including excavation, backfilling, compaction, and restoration of finished levels as per drawings and specifications.
- C. Construction of catch basins and kerb inlets at roads including earthworks, sand layers, plain concrete, reinforced concrete, and all civil works as per drawings and specifications.
- D. Box culvert structure including earthworks, foundations, reinforced concrete works, waterproofing, bases, walls, slabs, connections, all finishes, etc., complete as specified in authority specifications and drawings.
- E. Outfall structure including earthworks, foundations, reinforced concrete works, waterproofing, bases, walls (wing walls, head walls etc.) energy dissipater including grouted riprap, stone pitching, slabs, connections, all finishes, etc., complete as specified in authority specifications and drawings.
- F. All necessary works to complete the outfall structure including but not limited to excavation; dewatering as required, surface preparation, formwork, concrete and reinforcement, backfilling, stone pitching, concrete blinding, grating, waterproofing, protective coating, lining, associated equipment and accessories, and all other related works as per authority specifications.
- G. Pollution control devices including oil and debris interceptors and including all civil and mechanical work, installation, plan concrete reinforced concrete (if any), connection with pipelines and restoring finished surface.
- H. Open channel from the outfall to the pond location including slope protection, earthwork, excavation, backfilling, compaction as per drawings and authority's specifications.
- I. Ponds will be constructed including earthworks, channels, protections. all finishes, etc., complete as specified in authority specifications and drawings.
- J. Removal of existing storm network structures including manholes, pipes, catch basin, reinforced concrete works as per drawings.
- K. There are existing crossing MOT culverts along KKR & WRR have to be maintained
- L. Contractor shall coordinate with the local authorities / municipalities to obtain the construction approval before construction.
- M. The Contractor shall coordinate with all responsible authorities and ministries regarding data or location of all existing utilities and report to the Engineer.
- N. The Contractor will be responsible for the disconnection and reconnection of the existing services to the network.
- O. The Contractor shall prepare the shop drawings showing but not limited to pipe culverts, box culverts, pipeline route and sizes, catch basins, gullies, oil interceptors, ponds, outfalls all appurtenances and complete system as per authority's standards and specifications.

- P. The Contractor shall submit for Employer approval all technical information and samples of the materials he intends to use. These materials should be in accordance with the material specifications as per authorities and project specifications.
- Q. The Contractor shall commission the stormwater system upon completing the work and then shall be handed over to the authority.
- R. The Contractor will be responsible for the temporary diversion solution wherever required.
- S. Demolition works:
 - a. The Contractor shall ascertain the numbers and sizes of the pipes and culverts including pipe DIA, pipe material, quantities, and associated depth and location.
 - b. The Contractor shall be responsible for collecting the information with regards to existing network data from the relevant authorities.
 - c. The Contractor shall be responsible to submit to Employer and respective authority the demolition plan and schedule for approval before construction commencement.

6.10 Foul Sewer Network

- A. This section covers the performance of all the works and appurtenances required for foul sewer network and appurtenances as per the project drawings and other contract documents.
- B. Trial pits and survey data for manholes invert levels to be done to reconfirm line and level of existing manholes and pipeline to which connections will be made (if any).
- C. The scope shall include supply and installation of all pipework, including fittings, as per the drawings.
- D. Supply and installation of all manholes.
- E. Supply and installation of house connection chambers.
- F. Stub connections with end cap at plot limits for future connection by others.
- G. Installation of pipes and chambers both direct buried or within utility service culverts, if any, as per the drawings.
- H. Connection to already constructed / existing sewer lines, including connection from pressure network to gravity networks shall be completed in accordance with stakeholder requirements and tested accordingly.
- I. Testing and commissioning of all the new sewerage network.
- J. Disposal of surplus excavated material, waste materials, pipework and the like as directed by the Engineer.
- K. NWC General Specifications shall be followed. Where there is contradiction with the Technical Specifications, the Technical Specifications shall take precedence.

- L. Construction of one (1) foul sewage pumping station.
- M. Foul sewage pump station construction scope includes the following:
 - a. All civil, structural and architectural works shown on the pump stations drawings and included the pump stations contract specifications.
 - b. All mechanical, electrical, instrumentation, control and automation (MEICA) works shown on the pump stations drawings and included in the pump stations specifications.
 - c. Supply, installation, testing and commissioning of all MEICA equipment.
 - d. All works required to complete the scope of work to deliver fully functional system in accordance with the scope of work and Employer requirements.
 - e. Interface and connection with the potable water, irrigation networks, power networks, drainage networks, telemetry networks and SCADA systems as specified in the drawings and in the specifications.
- N. Demolition works:
 - a. Diversion, relocation, modification, or removal of already constructed networks where required as per the drawings.
 - b. The Contractor shall ascertain the numbers and sizes of the network to be demolished including pipe DIA, pipe material, quantities, and associated depth and location.
 - c. The Contractor shall be responsible for collecting the information with regards to existing network from the relevant authorities.
 - d. The Contractor shall be responsible to submit to Employer and respective authority the demolition plan and schedule for approval before construction commencement.

6.11 Irrigation Network

- A. This section covers the performance of all the work and appurtenances required for irrigation network as per the project drawings and other contract documents.
- B. Trial pits to reconfirm line and level of existing irrigation water pipelines to which connections shall be made at the MISK area as well as at the connection from DG Phase I area.
- C. Supply and installation of all pipework, including fittings, as per the drawings.
- D. Supply and installation of all chambers and valves as per drawings.
- E. Supply and installation of marker tape (or other agreed marker).
- F. Supply and installation all types of valves & chambers including but not limited to direct buried gate valves, gate valves in chambers, butterfly valves, air valves, washout valves, smart flow meters.

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- G. Connection to existing irrigation water pipelines at MISK area and connection of source from DGI existing / proposed pipeline from the proposed STP.
- H. Supply and installation of smart irrigation water meters as per authority's specifications to be compatible with authority's operation and control system including local and centralized SCADA system
- I. Testing and commissioning of all the new irrigation water network in accordance with the authority standards.
- J. Disposal of surplus excavated material, waste materials, pipework and the like as directed by the Engineer and in compliance with authorities' rules and regulations.
- K. Obtaining all approvals required from local pertinent authorities Where there is contradiction with the General and Particular Specifications, the Particular Specifications shall take precedence.
- L. Automatic Irrigation works shall include:
 - a. The Contractor shall prepare the shop drawings showing but not limited to The Main line route and sizes, Laterals route and sizes, gate valves, air valves, washout valves, risers, point of connections, etc. This design and construction scope includes primary irrigation network, while secondary (automated) irrigation will be provided by others and out of scope of the present contract.
- M. The Contractor shall submit for Employer approval all technical information and samples of the materials he intends to use. These materials should be in accordance with the material specifications as per authorities and project specifications.
- N. The Contractor shall commission the irrigation system upon completing the work and then shall be handed over.
- O. The Employer prior to issuing the project handing over certificate shall approve the As-Built drawings and the operation & maintenance manual.
- P. Demolition works:
 - a. Diversion, relocation, modification or removal of existing networks where required as per the drawings.
 - b. The Contractor shall ascertain the length of the irrigation network including pipe DIA, pipe material, valves, tanks (if any) and accessories, quantities and associated depth and location.
 - c. The Contractor shall be responsible for collecting the information with regards to existing irrigation network from the relevant authorities.
 - d. The Contractor shall be responsible to submit to Employer and respective authority the connection/disconnection plan and schedule for final approval before construction commencement.

6.12 Potable Water and Firefighting Network

- A. This section covers the performance of all the work and appurtenances required for potable water and firefighting network and relevant appurtenances as per the project drawings and other contract documents.
- B. Trial pits to reconfirm line and level of existing potable & firefighting pipelines to which connections shall be made.
- C. Supply and installation of all pipework, including fittings, as per the drawings.
- D. Supply and installation of all chambers and pillar type fire hydrants with protective barrier where directed.
- E. Supply and installation of marker tape (or other agreed marker).
- F. Supply and installation all types of valves & chambers including but not limited to direct buried gate valves, gate valves in chambers, butterfly valves, air valves, washout valves, smart flow meters
- G. Supply and installation of smart water meters as per NWC specifications to be compatible with NWC's operation and control system including local and centralized SCADA system.
- H. Connection to already constructed potable & firefighting pipelines.
- I. Diversion, relocation or removal of already constructed networks where required as per the drawings.
- J. Testing and commissioning of all the new potable & firefighting network.
- K. Disposal of surplus excavated material, waste materials, pipework and the like as directed by the Engineer.
- L. Construction of two (2) potable water and firefighting pumping stations.
- M. Pump stations construction scope includes the following:
 - a. All civil, structural and architectural works shown on the pump stations drawings and included the pump stations contract specifications.
 - b. All mechanical, electrical, instrumentation, control and automation (MEICA) works shown on the pump stations drawings and included in the pump stations specifications.
 - c. Supply, installation, testing and commissioning of all MEICA equipment.
 - d. All works required to complete the scope of work to deliver fully functional system in accordance with the scope of work and Employer requirements.
 - e. Interface and connection with the potable water, irrigation networks, power networks, drainage networks, telemetry networks and SCADA systems as specified in the drawings and in the specifications.

- N. Obtaining all approvals required from local authorities Specifications of General Directorate of Civil Defense Saudi Arabia – Fire Fighting and Requirements for protection from fire shall be followed. Where there is contradiction with General Specifications, the Particular Specifications shall take precedence.
- O. Contractor shall coordinate with the local authorities / municipalities to obtain the construction approval before construction.
- P. The Contractor shall coordinate with all responsible authorities and ministries regarding data or location of all existing utilities and report to the Engineer
- Q. The Contractor shall be responsible for the disconnection and reconnection of the existing services to the network.
- R. The Contractor shall prepare the shop drawings showing but not limited to pipeline route and sizes, valves, and complete system as per NWC standards and specifications.
- S. The Contractor shall submit for the Employer approval all technical information and samples of the materials he intends to use. These materials should be in accordance with the material specifications as per authorities and project specifications.
- T. The Contractor shall commission the potable and firefighting system upon completing the work and then shall be handed over.
- U. Demolition works:
 - a. Diversion, relocation or removal of already constructed networks where required as per the drawings.
 - b. The Contractor shall ascertain the length of the potable water network including pipe DIA, pipe material, valves including chambers, tanks and accessories, quantities and associated depth and location.
 - c. The Contractor shall be responsible for collecting the information with regards to existing potable water network from the relevant authorities.
 - d. The Contractor shall be is responsible to submit to Employer and respective authority the connection/disconnection plan and schedule for final approval before construction commencement.

6.13 Security Systems

- A. This section covers the performance of all the works and appurtenances required for security systems works and appurtenances as per the project drawings and other contract documents.
- B. The Contractor shall refer to a separate particular specification for detailed specifications of “Security Systems”
- C. Security monitoring and control system including:

- a. Unified Security System
 - b. Electronic Access Control System
 - c. Video Surveillance System
 - d. Automatic Number Plate Recognition System
 - e. Intruder Detection System
 - f. Perimeter Intruder Detection System
 - g. Security Network Communication
 - h. Security Intercom System
 - i. Vehicle Access Control
 - j. Guard huts and radar
 - k. Security Communications.
 - l. Roadways, and parking lots monitoring.
 - m. Authority requirements such as, but not limited to – Ministry of Interior (MOI).
- D. All responses to the tender shall be based on the contents of the security specification and associated security system drawings and relevant local authorities and Employer requirements.

6.14 Gas Network

- A. This section covers the performance of all the work and appurtenances required for gas network and relevant appurtenances as per the project drawings and other contract documents.
- B. Supply and installation of all pipework, including fittings, as per the drawings.
- C. Supply and installation of all chambers with protective barrier, where directed.
- D. Supply and installation of marker tape (or other agreed marker).
- E. Supply and installation all types of valves & chambers including but not limited to strategic ball valves, smart gas meters, etc.
- F. Testing and commissioning of all the gas network.
- G. Disposal of surplus excavated material, waste materials, pipework and the like as directed by the Engineer.
- H. Construction of one (1) SNG Gas station.
- I. SNG Gas station construction scope includes the following:
 - a. All civil, structural and architectural works shown on the gas station drawings and included in the contract specifications.

- b. All mechanical, electrical, instrumentation, control and automation (MEICA) works shown on the gas station drawings and included in the specifications.
 - c. Supply, installation, testing and commissioning of all MEICA equipment.
 - d. All works required to complete the scope of work to deliver fully functional system in accordance with the scope of work and Employer requirements.
 - e. Interface and connection of the plot with the potable water, irrigation networks, power networks, drainage networks, telemetry networks and SCADA systems as specified in the drawings and in the specifications.
- J. Obtaining all approvals required from local authorities Specifications of General Directorate of Civil Defense Saudi Arabia – Fire Fighting and Requirements for protection from fire shall be followed. Where there is contradiction with General Specifications, the Particular Specifications shall take precedence.
- K. Contractor shall coordinate with the local authorities / municipalities to obtain the construction approval before construction.
- L. The Contractor shall coordinate with all responsible authorities and ministries regarding data or location of all existing utilities and report to the Engineer
- M. The Contractor shall prepare the shop drawings showing but not limited to pipeline route and sizes, valves, and complete system as per authority standards and specifications.
- N. The Contractor shall submit for the Employer approval all technical information and samples of the materials he intends to use. These materials should be in accordance with the material specifications as per authorities and project specifications.
- O. The Contractor shall commission the gas system upon completing the work and then shall be handed over.

6.15 Pedestrian Underpasses / Box Structures

- A. This section covers the performance of all the works and appurtenances required for pedestrian underpass works and appurtenances as per the project drawings and other contract documents.
- B. The works include construction of two (2) road crossing structures (reinforced concrete box structures), earthworks, excavation, waterproofing, backfilling, compaction and restoration of finished levels as per drawings and specifications.
- C. Setting out of structure shall be performed in accordance with the drawings provided.
- D. Box structure including earthworks, foundations, concrete surrounds, reinforced concrete works, waterproofing, bases, walls, slabs, connections, all finishes, etc., to be completed as specified in authority specifications and drawings.
- E. Wingwalls/Retaining walls at the entrance and exit of the structures shall extend and tie-in to the toe line of road embankment.

- F. Contractor shall coordinate with the local authorities / municipalities to obtain the construction approval before construction.
- G. The Contractor shall coordinate with all responsible authorities and ministries regarding data or location of all existing utilities and report to the Engineer.
- H. The Contractor will be responsible for the disconnection and reconnection of the existing services to the network.
- I. The Contractor shall prepare the shop drawings showing but not limited to structures, all appurtenances and complete system as per authority's standards and specifications.
- J. The Contractor shall submit for the Employer approval all technical information and samples of the materials intended for use. These materials should be in accordance with the material specifications as per authorities and project specifications.
- K. The Contractor shall commission the structures and associated works upon completing the work and then shall be handed over.

6.16 Sustainability & Environmental Requirements

This section covers the performance of all the work and appurtenances required for sustainability framework compliances i.e., LEED Cities and Communities (LEED C+C_v4.1) and Mostadam Communities guide network as per the project drawings and other contract documents, in addition to Environmental requirements to be adhered to during construction.

- A. Contractor shall comply with the Diriyah Gate II Environmental Impact Assessment (EIA), the Outline Construction Environmental Management Plan (OCEMP), Environmental Protection and Enhancement Plan (EPEP) prepared and approved by DGCL in 2022, and LEED C+C_v4.1 Natural Resources Conservation and Restoration credit requirements.
- B. Contractor shall provide the below mentioned documentation list at the commencement of the Construction to Employer and Consultant for review and approval:
 - a) Contractor shall ensure that a scope-specific Construction Environmental Management Plan (CEMP) is developed and implemented including but not limited to the requirements listed in the Diriyah Gate II OCEMP. The CEMP shall be developed in accordance with the requirements outlined in the OCEMP and is to be evaluated and approved by a qualified Environmental Professional.
 - b) Contractor to ensure the CEMP includes the Environmental Management Plans and Sustainability Management Plan as stipulated in the Diriyah Gate II OCEMP. These plans include but are not limited to:
 - i. Construction Activity Pollution Prevention Plan including Erosion & Sedimentation Control Plan. The plan must conform to the erosion and sedimentation requirements of the 2017 U.S. Environmental Protection Agency (EPA) Construction General Permit (CGP) or local equivalent, whichever is more stringent. Projects must apply the CGP regardless of size.
 - ii. The Contractor shall develop and implement a scope-specific Construction & Demolition Waste Management Plan in line with the outlined CDWMP (doc ref DG220-

JCB-DE-TCR-000090) complying with the waste diversion rate of 70% (excluding hazardous waste) from landfill as per the LEED C+C_v4.1 Construction and Demolition Waste Management credit requirements.

- c) Contractor to ensure the CEMP includes an Environmental Monitoring Programme. The programme is to be developed and implemented as per the requirements listed in the Diriyah Gate II OCEMP.
 - d) Sustainability Action Plan (LEED C+C_v4.1 & Mostadam Communities).
 - e) Sustainability Management Plan.
 - f) Commissioning Plan.
 - g) Commissioning Report including.
 - iii. Installation Checklist.
 - iv. Pre-functional Checklist.
 - v. Functional Checklist.
 - vi. Commissioning Checklist.
 - vii. Functional Performance Test Results.
 - viii. Testing and Balancing Report.
 - h) List of Material with clearly identified materials applicable for LEED C+C and Mostadam Communities requirements.
- C. The Contractor shall comply with the vulnerability and resilience plan as per LEED C+C_v4.1 Resilience Planning credit requirements.
- D. The Contractor shall provide proper number of Electric Vehicle parking, Signages and EVSE (Electric Vehicle Supply Equipment) as per below LEED requirement:
- Provide a Level 2 charging capacity (208 – 240 volts) or greater.
 - Comply with the relevant regional or local standard for electrical connectors, such as SAE Surface Vehicle Recommended Practice J1772, SAE Electric Vehicle Conductive Charge Coupler or IEC 62196 of the International Electrotechnical Commission for cities outside the U.S.
 - Be capable of responding to time-of-use market signals (e.g., price). Cities pursuing EN credit Grid Harmonization should incorporate EVSE into any demand response program or load flexibility and management strategies.
- E. The Contractor shall comply with all the requirements for LEED C+C_v4.1 Smart Water Systems such as smart water meter installation for 100% of water connection units.
- F. The Contractor shall install all street lighting and public area lighting in compliance with 'ANSI/IESNA RP-8-14 Roadway Lighting' or internal equivalent standard. In addition, lamps shall have a minimum Luminous Efficacy of 125 lumens per watt.

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- G. The Contractor shall install all water and wastewater pumps in compliance with Pump Energy Index listed in Table I.1—Proposed Energy Conservation Standards for Pumps, 10 CFR Parts 429 and 431 of DOE standards Federal Register final rule Energy Conservation Program: Energy Conservation Standards for Pumps or international equivalent standard.
- H. The Contractor shall ensure that at least 40% (by cost) of permanently installed top five infrastructure materials comply with the one of the following as per LEED C+C_v4.1 Responsible Sourcing for Infrastructure credit requirements.
 - a. Material Reuse - Reuse includes salvaged, refurbished, or reused materials/products.
 - b. Recycled content
 - c. Leadership extraction practices – USGBC approved program – Other USGBC approved program meeting responsible sourcing and extraction criteria.
- I. The Contractor shall comply with the metering strategy as per supporting guidance provided in Mostadam Communities E-01 Energy metering and W-01 Water metering and provide energy and water meters for all major end uses and sub-use types (i.e., minimum of one meter per system)
- J. The Contractor shall appoint a third-party Testing, Adjusting and Balancing contractor to commission the systems such as Pumps, lighting, and controls system as per Commissioning Plan develop by Independent Commissioning Authority as required by LEED Cities and Communities_v4.1 and Mostadam Communities credit requirements.

7. SITE CONDITIONS

7.1 Site Conditions

The following information is given solely for general assistance (Information and Inspection) the Conditions of Contract shall be applicable in relation thereto:

- A. Ambient Air Temperatures
 - a. Maximum 50 degrees C
 - b. Minimum 7 degrees C

- B. Relative Humidity

- a. Maximum 100%

- C. Rainfall

As is usual in areas of low precipitation, distribution of rainfall is very erratic. Many rainstorms cover relatively small areas so that it is difficult to obtain records of maximum intensities. Heavy showers occur.

- D. Fog and Dew

During certain periods of the winter month's night and early morning fog is common. This may persist from two hours after sunset until two hours after dawn. Similar conditions can also occur in early mornings in the summer months.

Heavy dew can be expected most mornings at any time of the year.

- E. Winds

The records give no information of long-term maximum gusts. Gusts of 23 m/s and 35 m/s respectively have been recorded at. The estimated wind speed for a 3 second gust, likely to be exceeded once in 50 years, is 44 m/s.

- F. Sandstorms

Sandstorms are occasionally experienced mainly during the months of January to March and August to October.

- G. Earthquakes

The area should be considered as an area of moderate intensity and susceptible to seismic activity of intensity of MMV II (Modified Mercalli Scale) or greater in a period of 60 years.

8. TEMPORARY SITE OFFICES

8.1 Temporary Offices and Associated Sanitary Facilities

The Contractor shall be responsible for and make its own arrangements for providing and equipping for its own and its Subcontractors' use all temporary offices and associated sanitary facilities including connections to services and sewers on the Site as may be required during the carrying out of the Works on Site in an area to be allocated by the Employer. All connections to services and sewers on the Site are subject to the Engineer's prior approval. The Contractor shall also provide utility, cleaning and maintenance services to such offices and associated sanitary facilities. All accommodation, equipment and facilities shall be of a good standard and shall be to the approval of the Engineer. All costs associated with providing and maintaining such temporary offices and facilities shall be borne by the Contractor.

8.2 Engineer's Site Office

The Contractor shall make furnished site office available for the Engineer's use prior to start of the Work at Site and to remain on Site for minimum of 30 days after final acceptance of the Work. The office shall be located where directed by the Engineer. Minimum features of the office shall be:

- A. Lighting and wall plugs.
- B. Wiring and equipment necessary for wireless connection for mobile phones and computers.
- C. Fluorescent ceiling lights.
- D. Self-contained air conditioning units properly sized for Project locale and conditions. Provide ample electric power to operate installed systems.
- E. Railed stairways and landings at entrances.
- F. Sign on entrance door reading Jacobs (CH2M), letter height 100mm minimum.
- G. Security guard screens on windows.
- H. Toilet and wash basin in separate compartment with hot and cold water and drains.
- I. All accommodation, equipment and facilities shall be of a good standard and shall be to the approval of the Engineer.

The Contractor shall maintain in good repair and appearance, and also provide utility, cleaning and maintenance services to such offices and associated sanitary facilities. This shall include replenishment, as required, of paper towels, paper cups, hand soap, toilet paper, first-aid kit supplies, fire extinguishers and bottled drinking water.

All Costs in connection with providing and maintaining such temporary offices and facilities shall be borne by the Contractor.

8.3 Engineer's Field Office Equipment

The Contractor shall provide equipment specified for exclusive use of Engineer and its representatives for the duration of the project. Ownership of equipment furnished under this article will remain, unless otherwise specified, that of the Contractor. Equipment furnished shall be new or like new in appearance and function and as a minimum shall consist of:

- A. Bottled water service: One, with cooler capable of producing hot water and cold water.
- B. Light refreshments: Tea/Coffee and bottled water including the daily service of an 'office boy'/cleaner.
- C. Paper cup dispenser with cups: One.
- D. Paper towel dispenser with towels: One.
- E. First-Aid Kit: One.
- F. Automatic External Defibrillator (AED): One
- G. Tri-Class (ABC), dry chemical fire extinguisher, 10pound: One.
- H. Printer/ plotter: One.
- I. Telephone: one per desk

All Costs in connection with installation, maintenance and supply of consumables shall be borne by the Contractor.

8.4 Temporary Services and Facilities

8.4.1 Electricity

The Contractor shall provide supplies of electricity required for the carrying out of the Works on Site and all costs in connection with such supplies shall be borne by the Contractor.

8.4.2 Water

The Contractor shall provide supplies of water required for the carrying out of the Works on Site and all Costs in connection with such supplies shall be borne by the Contractor.

8.4.3 Compressed Air

The Contractor shall provide all compressed air required for carrying out of the Works on Site and all Costs in connection with such provision shall be borne by the Contractor. Diesel engine driven compressors shall not be sited within buildings or in locations which may cause a health hazard to personnel due to exhaust fumes.

8.4.4 Medical Facilities

Contractor shall comply with labor and other applicable KSA Laws.

8.4.5 Temporary Telephones

The Contractor shall be responsible for and make its own arrangements for the supply and installation of any temporary telephone and internet facilities that may be required during the carrying out of the Works on Site and all costs in connection with such facilities shall be borne by the Contractor. Contractor shall coordinate with service providers for provision of temporary GSM tower for mobile coverage during the construction.

8.4.6 Storage, Fabrication and Assembly Area

The Employer will allocate to the Contractor an area for its exclusive use for storage, fabrication and assembly. The Contractor will be responsible for providing all buildings, structures, services and hard standings required by itself and its Subcontractors and shall erect a temporary security fence around the area.

8.4.7 Ablution Facilities

The Contractor shall be responsible for and make its own arrangements for providing for its own and its Subcontractors' use temporary ablution facilities on the Site as maybe required during the carrying out of the Works on Site in an area to be allocated by the Employer. The Contractor shall maintain these facilities in a clean and hygienic condition and shall be responsible for the emptying of any tanks or cesspits. All Costs in connection with providing and maintaining such ablution facilities shall be borne by the Contractor.

8.4.8 Design of Temporary Works

The temporary works shall be designed and constructed in such a manner as to enable the Works to be executed around them.

All services or utilities on or adjoining the Site which are required to remain operational during the carrying out of the Works shall be protected by the Contractor from movement, subsidence or damage from any cause whatsoever by adequate temporary props, struts, shores, protective screens and the like to the approval of the Engineer and all relevant Authorities.

8.4.9 Access

Access to and from the Site shall be either the Haul and site routes access points agreed with the engineer.

Access routes for construction traffic within the Site area shall be agreed with the Engineer before use but shall be in line with the logistics plan for construction activities. Should the Engineer require any widening, strengthening, repair, protection or alteration of any kind to access routes within the area of the Site, the Engineer may at any time withdraw approval for use of the said access routes until such widening, strengthening, repair, protection or alteration of any kind has been carried out to the satisfaction of the Engineer. All Costs of such widening, strengthening, repair, protection or alteration of any kind shall be borne by the Contractor.

Entrance to the project area will be subject to the Employer's security system for which temporary passes or permits may be required for the Contractor's personnel. All costs and requirements associated with obtaining such passes to be borne by the Contractor.

8.4.10 Lifting Equipment

All lifting equipment shall be provided by the Contractor, at its own Cost, and it shall be fully responsible for the operation and maintenance of such equipment. All slings and lifting equipment shall be proof load tested and stamped prior to use on the Site and shall be subject to approval by the Engineer, who has the right to prohibit the use of equipment or methods considered unsafe at his sole discretion.

The Contractor shall provide the Engineer with copies of the load test certificates for all lifting equipment brought onto the Site.

8.4.11 Protection

The Contractor shall take all precautions to protect the Works, workmen and the public and shall be responsible for providing, at its own cost, all necessary temporary screens, fencing, barriers and any other protective equipment and the like deemed necessary by the Engineer to his approval. The Contractor shall provide secure, safe and protected areas for visitors.

The Contractor shall, wherever required or wherever directed by the Engineer, cover up and protect the Works from the weather and from damage by any subsequent operations. The Contractor shall provide all necessary dust sheets, padding, protective strips, sheets, panels and layers, barriers, guard rails and the like and clear away the same at completion.

The Contractor shall protect to the maximum extent possible any and all places or materials on or about the Site which may be hazardous or dangerous to its workmen, other persons or to traffic. The Contractor shall provide and maintain warning and caution signs, warning lamps, barricades and any other protective equipment and the like deemed necessary by the Engineer to his approval.

8.4.12 Site Security

The Contractor shall be responsible for and make its own arrangements for providing all necessary site security arrangements that may be necessary during the carrying out of the works on Site and all costs in connection with such security provision shall be borne by the Contractor.

8.4.13 Messing Facilities

The Contractor shall be responsible for and make its own arrangements for providing all necessary messing and catering facilities that may be required for all grades of its own and its Subcontractors' staff and all costs in connection with such facilities shall be borne by the Contractor.

Should catering facilities be made available to the Contractor by the Employer arrangements for the use of these facilities shall be made directly with the Employer. All costs associated with using these facilities shall be borne by the Contractor.

8.4.14 Transportation of Personnel

The Contractor shall be responsible for and make its own arrangements for the transportation of all persons employed by the Contractor and its Subcontractors in connection with carrying out the works and all costs in connection with such transportation shall be borne by the Contractor.

The vehicles to be used for transportation of persons employed by the Contractor and its Subcontractors shall be subject to the prior approval of both the Engineer and the Employer. In this respect, suitable buses, coaches and the like will be considered but proposals to use personnel carriers and the like will not be acceptable.

8.4.15 Cleaning of Site

The Contractor shall maintain the Site in a neat and orderly condition, free from accumulation of waste materials and rubbish, during the carrying out of the Works, all to the satisfaction of the Engineer. Immediately prior to the handing over of the Works all drain pipes, ducts, gullies, manholes, chambers or any other openings are to be cleared by rodding or by other means approved by the Engineer.

The Contractor shall be responsible for and make its own arrangements for the clearing away and removal of all waste materials and rubbish from the Site at regular intervals during the progress of the Works and all Costs in connection with such clearing away and removal shall be borne by the Contractor.

8.4.16 Removal of Temporary Works

Upon completion of the Works the Contractor shall remove from the Site and any adjacent areas affected by the Works, all temporary works, debris and other works remaining thereon and all disturbed works and ground shall be made good all to the satisfaction of the Engineer.

8.4.17 Site Laboratory

The Engineer shall provide the Contractor with details of the laboratory tests, which are required to be carried out on Site. The Contractor shall provide, equip, operate and maintain a laboratory to the approval of the Engineer for all such tests. All costs in connection with such provision, equipping, operation and maintenance shall be borne by the Contractor.

8.4.18 Temporary Roads

The Contractor shall be responsible for the construction and maintenance of all temporary roads that will be required within the Site area during the carrying out of the works on site and all costs in connection with such temporary roads shall be borne by the Contractor. Temporary roads shall be positioned in accordance with the Engineer's written instructions and shall be kept free for access for all parties at all times. The Contractor shall be responsible for the implementation and maintenance of strict safety measures on all temporary roads in accordance with the Engineer's requirements.

9. INSPECTION AND TESTING BY THE ENGINEER DURING MANUFACTURE

As part of the requirements of the Contract, the following procedures shall be adopted. All references to tests are deemed to include inspection, examination and/or testing as appropriate during manufacture.

After the Commencement Date the Engineer may review the Contractor's and its Subcontractors' methods of quality control to ascertain that:

1. The inspection and testing programme during manufacture contains all the tests and inspection stages required by the terms of the Specification; and
2. The quality control procedures during manufacture are adequate in scope and detail and that the manufacturer's inspection and testing programme is competently implemented and documented.

During the course of the Contract, the Engineer will issue to the Contractor a statement giving details of the manufacturing stages and tests which the Engineer will wish to witness.

During each visit to the Works where the testing is to be carried out the Contractor shall permit the Engineer to survey the quality control systems relating to the item of plant, equipment or work element subject to test.

The Contractor shall inform the Engineer formally at least ten days in advance of the date when any items of plant, equipment or other work element will be ready for testing, stating the following:

1. A description of the item to be tested.
2. The reference to the control stage given in the Engineer's statement referred to above.
3. The name and address of the place at which the Plant will be tested and the name of the responsible person at that address.
4. The date on which the Plant will be ready for testing.

The Engineer will notify the Contractor formally at least twenty-four hours before the due date for the tests if he intends to be represented at the tests. A record of all test results shall be handed to the Engineer on completion of his inspection visit. Within fourteen days from receipt of approval of the test results, the Contractor shall issue four copies of the official test certificates to the Engineer.

When the Engineer is satisfied that the item of Plant has satisfactorily passed its tests and complies with the Specification and the Contract Drawings and decides that no further inspection or testing prior to delivery is necessary, he will issue to the Contractor a Release certificate to signify that the Plant has passed the Engineer's inspection control during manufacture.

If the Engineer does not attend the tests the Contractor shall apply to the Engineer for confirmation that the relevant Plant has passed the inspection control at the manufacturer's works. Within fourteen days of the receipt of this application and of satisfactory test results the Engineer will issue the said confirmation in the form of a Release Certificate.

Should however, during the course of the Works any item of Plant be found not to be in accordance with the Specification and the Contract Drawings, the Contractor will be held responsible for making the necessary alteration or replacement of the defective item at its own Cost to the satisfaction of the Engineer.

10. TRANSPORT AND HANDLING

10.1 Transportation to Site

The Contractor shall arrange for and bear and pay all Cost in connection with the delivery to the Site of all Plant and Contractor's Equipment.

The Contractor shall obtain the Employer's permission, giving at least forty-eight hours' notice, before delivering any large or heavy loads to the Site. Axle loads shall not exceed nineteen tonnes.

10.2 Storage, Handling and Unpacking

The Contractor shall be responsible for the off-loading, handling, storage and unpacking of all Plant and Contractor's Equipment. The Contractor shall submit the method statement for the Engineer's approval prior to the commencement.

The provision of any covered stores or preparation of hard standing storage areas shall be the Contractor's responsibility.

10.3 Protection of Plant

The Contractor shall arrange for the adequate protection of all Plant items against deterioration, corrosion or damage during storage, erection, testing and commissioning at the Site

Whenever possible the delivery of switchgear, indoor instruments and panels shall be arranged direct to their respective places of installation. Such items shall not be unpacked or stored in a non-air-conditioned environment.

During the storage time and the period subsequent to erection but prior to commissioning, motors, transformers and other electrical apparatus fitted with anti-condensation heaters which are subjected to the prevailing ambient conditions shall have their anti-condensation heaters connected to a temporary or permanent power supply to prevent deterioration of the electrical insulation. Such connection shall be carried out by the Contractor whether they are in the storage area or in their permanent locations.

11. SITE REQUIREMENTS

11.1 Setting-Out and Levels

The Contractor shall give the Engineer at least twenty-four hours' notice of its intention to set-out or fix levels for any part of the Works so as to enable the Engineer to arrange for checking. The Contractor shall employ competent surveyors and assistants to ensure the correct setting out of the Works in accordance with the Contract Drawings. Records of setting-out and levels taken shall be made and submitted to the Engineer.

11.2 Use and Protection of the Site

The Contractor shall take all necessary measures and exercise all necessary care in order to properly protect the Site, and the Works during the carrying out of the Works all to the satisfaction of the Engineer.

The Contractor shall take all measures necessary to confine its activities to and prevent all personnel employed directly or indirectly by it from trespassing beyond the area which is allocated to it and shall indemnify the Employer against all claims arising in respect of trespass.

Subject to any provision to the contrary contained in the Contract, the Contractor shall not disturb, damage or pull down any hedge, bush, tree, wall or building on the Site without the prior approval of the Engineer.

The Contractor shall not use the Site for any purpose other than carrying out the Works.

11.3 Contractor's Working Area

An area will be allocated to the Contractor as a working area within which he must confine such operations and temporary buildings as may be required to complete the Works. Throughout the period of the carrying out of the Works, the Contractor shall maintain the whole area of its operations in a clean, tidy, secure and safe condition.

All rubbish, waste material, debris and the like shall be systematically cleared off the working areas as it accumulates and if not removed directly from the Site shall be deposited at collecting points allocated by the Engineer pending removal from the Site.

11.4 Name Boards and Advertisements

The use of any advertisements press releases, publicity photographs and the like relating to or including reference to the Site, the Works, the Contract or the Employer is subject to such use being approved by the Employer in writing prior to publication.

11.5 Diversion of Traffic

The Contractor shall construct temporary diversion Works in accordance with MOT specification, wherever the execution of the Works will interrupt existing roads.

Prior to commencing any individual diversion or traffic control, the Contractor shall submit to the Engineer all details of his proposed traffic control plan through a third-party Traffic Consultant and approved by the Engineer, including all detours, layouts, geometrics, pavement construction, drainage, lighting, signing, traffic control

devices including pavement marking and existing marking removal, crashworthy safety barriers according to MOT Specification. The Contractor should provide to the Engineer his programme of diversion works for the entire project. All diversions must be approved by the Engineer, the Traffic Police and MOT prior to implementation. The Contractor has to conduct Road Safety Audits (RSA) for the temporary diversions through a third-party RSA Consultant and approved by the Engineer. The Contractor shall submit the final RSA report to the Engineer for approval.

11.6 Obstruction of Access

The Contractor shall take precautions to keep all roadways clear, to the satisfaction of the Engineer, of any spillage from its traffic. All such spillage which occurs shall be cleared immediately to the Engineer's satisfaction and at the Contractor's Cost. Spillage shall, without limitation, include excessive earth brought in on wheels or tracks of traffic.

11.7 Safety, Health and Welfare

When working on the Site, any other area for the time being occupied by the Employer or any area designated by the Engineer, the Contractor shall at its own Cost strictly observe and perform each and every provision of the Employer's rules, regulations and procedures relating to safety, health and welfare for the time being in force.

Smoking shall be limited to such locations and occasions as are specifically authorized by the Employer. Safety clothing is to be worn and safety equipment is to be used by the Contractor and all personnel for whom it is responsible.

The Contractor shall in respect of safety, health and welfare matters abide by the applicable laws and regulations of the Kingdom of Saudi Arabia (KSA).

The Contractor shall maintain such records and make such reports concerning safety, health and welfare of persons, and damage to property as the Engineer or the Employer may from time to time require.

The Contractor shall report to the Employer and the Engineer details of any accident or LTI as soon as possible after its occurrence. The Contractor shall in addition report all accidents to the appropriate authorities to the extent required by law. In the case of any fatality or serious accident, the Contractor shall, in addition, notify the Employer and the Engineer immediately by the quickest available means.

11.8 Precautions Against Fire

The Contractor shall comply with the following requirements:

1. Provide adequate firefighting appliances at the Site.
2. Maintain proper storage and handling of flammable liquids and gases.
3. Maintain good housekeeping standards including safe disposal of all scrap material.
4. Restrict smoking by workmen to authorized safe areas.
5. Maintain access routes for emergency response vehicles.
6. Provide adequate fusing protection and guarding of temporary electrical installations and provide correct and proper connections to the Employer's electrical equipment.

The Contractor shall use all reasonable endeavours to ensure that all of its employees and its Subcontractors' employees take all reasonable precautions during and outside normal working hours to prevent the outbreak of fire. The Contractor shall prepare the firefighting strategy which includes, but not limited to, emergency exit plan,

firefighting equipment list and location, fire warden details, assembly point etc. in accordance with the Civil Defence guidelines and submit to the Engineer for approval.

The Contractor shall follow the Employer's fire and safety procedures at all times.

11.9 Working Hours

Refer to the Conditions of Contract for details regarding working hours.

11.10 Scaffolding

The supply, erection, maintaining, dismantling and clearing away of all scaffolding work shall be carried out by a specialized scaffolding Subcontractor and all such scaffolding shall be supplied and erected in accordance with BS 5973 and BS 5974. The said Subcontractor shall be subject to the prior written approval of the Employer and the Engineer.

Any holes made for the scaffolding work shall be made good to match the adjacent surfaces after the scaffolding is dismantled. The Contractor shall be entirely responsible for all safety precautions in connection with scaffolding work.

11.11 Existing Site Services

The location of existing services shall be verified as far as practicable by the Contractor on the Site and the Contractor shall be responsible for the protection of all existing services and other Works.

Excavation in the vicinity of known services shall be by hand and all known buried services must be located by hand excavation before machine digging is permitted. No machine excavation shall be allowed when excavating above existing cables/utilities. The Contractor shall immediately reinstate at no additional Cost to, and to the satisfaction of the Employer, any service that may be severed or disturbed. The Employer may, if it deems fit, engage the services of others to reinstate any service severed or disturbed by the Contractor and the Cost of such reinstatement as determined by the Engineer shall be recoverable from the Contractor by the Employer as a debt due from the Contractor, or may be deducted by the Employer from any monies due or which may become due to the Contractor.

The Contractor shall liaise with all service providers to confirm and identify all existing services information. In the event of unknown services being discovered during the carrying out of the Works, the Contractor shall immediately inform the Engineer in writing. Payment for diverting or sealing off and abandoning any of these services if necessary or working around the same will only be made where the proviso to (Information and Inspection) the Conditions of Contract applies and even then, only against certified records and subject to the prior authorization of the Engineer with the consent of the Employer. The Contractor shall locate dimensionally on its as-built drawings all services encountered on the Site

11.12 Work by Others

The Contractor is obliged to interface with other contractors on the site including contractors working on adjacent packages for proper interface and alignment of services at the tie-in connection points. This interface shall be

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deemed to include a minimum of attending coordination meetings as well as the provision of relevant information between contracting parties to enable the timely completion of other contracts.

The sourcing of the relevant information required to produce these Shop Drawings shall additionally be deemed part of the Contractor's program obligations.

Where any part of the Works depends for its proper performance and execution upon the Work of any other Contractor, the Contractor, prior to carrying out such part of the Works, shall inspect and promptly report to the Engineer any defect or deficiency in such Work of any other Contractor that renders it unsuitable for such proper performance and execution. If the Contractor fails to so inspect and report any such defect or deficiency and proceeds with the carrying out of the said part of the Works, such carrying out shall be deemed to be an acceptance by the Contractor of the said Contractor's Work for the purposes of carrying out the Works.

As part of ensuring the proper performance and execution of the said part of the Works, the Contractor shall prior to commencing such part of the Works measure the said Work of the other Contractor and shall at once report to the Engineer any discrepancy between such Work and the relevant Drawings.

12. TESTING AND COMMISSIONING

12.1 Test Certificates

The Contractor shall submit details of its proposed testing and commissioning procedures and check lists to the Engineer at least two months prior to the start of the Tests on Completion for any Section or Portion of the Works.

Within two months of the completion of the Tests on Completion and commissioning for any Section or Portion of the Works the Contractor shall submit a comprehensive test and commissioning report with bound sets of completed check sheets as follows:

- A. one copy to the Engineer
- B. six copies to the Employer

Within one month of completion of the tests provided for in the Conditions of Contract the Contractor shall submit a comprehensive test report with the same distribution as the test and commissioning report.

12.2 Operations and Maintenance Manuals

12.2.1 Requirements for Manuals

The Contractor shall provide Operations and Maintenance Manuals for each section and portion of the Works. It shall be a condition precedent to the issue of the Taking-Over Certificate under the Contract for any Section or Portion of the Works that the Contractor shall have delivered to the Engineer the Manuals relating to that section or portion of the Works.

The page size shall be standardized for all Manuals to A4 international standard and shall be punched to fit standard four-hole fixings. The use of fold-out pages or turn-pages shall be kept to a minimum. Fold-out pages shall be held to a two-page fold-out wherever practicable and shall show the material thereon with the Manuals closed. Each Manual volume shall be collated in its proper order and placed in a black durable rigid cover with inlaid gold lettering. The binder shall be of the type that may be taken apart without the use of threaded binding posts.

The contents of the Manuals shall conform with a Table of Contents to be included in each Manual which shall be complete and specific. Only material specific to the Contract shall be used. Nomenclature or reference to any one item shall be consistent throughout the Manuals. The information provided shall be complete for both main and auxiliary equipment and systems provided by the Contractor. Material that does not contribute to the understanding of the design, operation and maintenance of the equipment or system shall be excluded from the Manuals.

Use shall be made of Drawings, diagrams, pictures or actual photographs when they add to the understanding and clarity of the text. The Manuals are to be produced by photostatic, lithographic or similar processes. The use of coated papers and dyeline prints for Drawings or script within the Manuals is unacceptable. All material shall be free from stamps commonly used for identification of the customer, order number, and the like.

Precautions and warnings relative to the safety of life and equipment shall be included.

12.2.2 Arrangement and Format

The Manuals shall be divided into volumes covering each part of the Works. The first volume shall include the following:

- Title page.

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- Index of volumes and comprehensive reference to other volumes.
- Table of contents.
- Description of how to use the Manuals and introduction.
- Overall operating philosophy.
- Operating conditions.
- General description of equipment and controls.
- Description of operation of the Plant.
- Plant arrangement Drawings.

Volumes shall be provided for each part of the Works and shall include the following:

- Title page.
- Table of contents.
- Isometric Drawings, pictures or photographs of major items.
- Section I - Operation, including:
 - a. General description
 - b. Performance specifications
 - c. Starting instructions
 - d. Operating instructions
 - e. Shut-down instructions
 - f. Emergency procedures
 - g. Preservation instructions
 - h. Installation and commissioning instructions
 - i. Design data and curves.

This section of the Manuals shall contain:

General description of all equipment including overall design and specific and special features of design including descriptive Drawings where practical.

Technical particulars and performance particulars for all equipment.

Complete, detailed and specific operating instructions for all equipment provided. Included shall be precautions and critical points to be observed including suggested forms to be used in taking periodic readings to maintain an operations record. These shall include a tabulation of possible operation difficulties with the probable causes listed and remedial action to be taken under each one.

Complete, detailed and specific shut-down instructions for all equipment provided, noting the step-by-step procedure to be followed for shutting down the equipment. Precautions and critical points to be observed shall be noted and emphasized. These Instructions shall be divided into 'Normal Shut-Down' and 'Emergency Shut-Down'.

Preservation instructions detailing procedures and precautions to be taken in the event of any Plant being shut-down for extended periods.

Design data for all equipment and systems specifying power, voltage, current, pressure, temperature, revolutions per minute, flow and the like.

- Section II -Maintenance, including:
 - a. Fault finding, diagnosis and rectification` -Dismantling instructions
 - b. Maintenance instructions -Settings, clearance and adjustment data -Assembly instructions - Circuit diagrams -Exploded views.
- Section III -Parts Catalogue, including:
 - a. A comprehensive list of all replacement parts
 - b. Drawings and lists
 - c. Instructions for ordering replacement parts.

This section of the Manuals shall contain:

Replacement parts, Drawings and lists completely detailed and with specific replacement part Drawings and lists for all equipment assemblies and sub-assemblies. The material shall cover all information required for ordering replacement parts such as part name, part number, equipment serial number, original manufacturer and the like.

Complete instructions for ordering replacement parts in a manner that would prevent errors or misunderstandings. Recommended forms for tabulating replacement part information and instructions for returning material to the factory shall also be included. Special storage or handling procedures required for any particular parts shall be noted.

Separate volumes shall be provided containing all test certificates and test reports for the Works. These shall be fully indexed, and cross referenced to the other volumes.

12.2.3 Submission

The Contractor shall prepare and submit one draft copy of the Manuals to the Engineer and one draft copy of the Manuals to the Employer at least three months before the commencement of the Tests on Completion. These draft Manuals shall be sufficiently comprehensive for the Employer's and the Engineer's use during the Tests on Completion, commissioning, initial operation and for training purposes.

The Engineer's and the Employer's comments as a result of the Tests on Completion, commissioning and initial operation shall be given by the Engineer to the Contractor within three months of the completion of the Tests on Completion. The Contractor shall revise the Manuals as necessary upon receipt of the Engineer's and the Employer's comments and submit two hard copies of such revised Manuals to the Engineer for approval by the Engineer and the Employer.

Upon receipt of a notification from the Engineer confirming the approval of the Engineer and the Employer to the revised Manuals, the Contractor shall prepare and submit to the Engineer for the approval of the Engineer and the Employer such revised Manuals as per the submission procedure mentioned in the Employer's Specification or otherwise instructed by the Engineer.

12.3 Computer Software

The Contractor and its Subcontractors shall comply with all applicable rules, regulations and procedures of the Employer relating to computer software or firmware brought onto the Site. In particular, all computer software or firmware brought onto the Site shall be clear of viruses or like agents. The Employer shall be entitled at any time

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to test such firmware or software for viruses or like agents. All software and firmware provided by the Contractor under the Contract shall be compliant with the Employer requirements.

13. COORDINATION

13.1 General

- The Contractor shall coordinate the various parts of the Works in an efficient and orderly manner.
- The Contractor shall be responsible for coordinating all Drawings including those updated by the Consultant and those produced by Subcontractors, Government and Municipal Services Authorities or others employed directly in connection with the Works.
- The Contractor shall employ a competent person experienced in the coordination of engineering and other installations to perform this duty. He shall arrange for its Subcontractors, Government and Municipal Services Authorities or others employed direct to attend the necessary coordination meetings.
- Any amendments to the Drawing's consequent upon the coordination of the various installations shall be made by the Subcontractors or others concerned prior to such Drawings being submitted to the Engineer for approval.
- The Contractor shall be responsible for ensuring that Subcontractors and others programme their Work and provide the Contractor with information regarding the Work in reasonable time relative to its own programme.
- The Contractor shall coordinate space requirements and installation of mechanical and electrical work and utilize spaces efficiently in order to maximize accessibility for other installations, for maintenance and for repairs.
- Upon award of the Contract, the Contractor has to prepare and submit a detailed program showing diversion/protection of all existing utilities including the overhead power lines to meet the construction completion date for the review and approval of the Engineer.
- The Contractor shall coordinate and liaise with other contractors or Tenders which will be issued by the Employer during the course of Works.

14. EXISTING UTILITY PROTECTION / DIVERSION WORKS

It is the responsibility of the Contractor to obtain all the relevant NOCs and existing information from all the authorities to identify all the existing services in the project area. The Contractor has to undertake the relevant trial pits to locate the type, depth and other details of the service after obtaining consent from the relevant authority. All the affected services are to be protected or relocated to the proper corridor which has been identified in the relevant drawings and documents under the tender submission. The Contractor needs to further identify during the construction stage and prepare relevant Shop Drawings in coordination with the relevant authorities and the Engineer on Site to obtain all necessary approvals. The existing services identified during the design stage and responses received from service authorities are included in the Contract Drawings, however, the Contractor shall approach all the authorities to verify the information provided and also to identify all services in the work area and propose the relevant protection/diversion methods in coordination with the authority and Engineer on site. The utility protection/diversion works are to be constructed in accordance with the relevant authority standards.